



AGRICULTURE — CLASS 10 GEOGRAPHY

Complete NCERT-Based Notes

1. AGRICULTURE — INTRODUCTION

- India is an agriculturally important country.
- **Two-thirds** of its population is engaged in agricultural activities.
- Agriculture is a **primary activity**, which produces most of the food that we consume.
- Besides food grains, agriculture also produces **raw material** for various industries.
- Some agricultural products like tea, coffee, spices, etc. are also exported.

2. TYPES OF FARMING

- Agriculture is an age-old economic activity in our country.
- Over the years, cultivation methods have changed significantly depending upon:
 - **physical environment**
 - **technological know-how**
 - **socio-cultural practices**
- Farming varies from subsistence to commercial type.
- At present, in different parts of India, various farming systems are practised.

Slash and Burn Agriculture

- Farmers clear a patch of land and produce cereals and food crops.
- When soil fertility decreases, farmers **shift and clear a fresh patch of land**.
- Allows Nature to replenish fertility through natural processes.
- Land productivity is **low** (no fertilisers or modern inputs used).

Names of Shifting Cultivation

Region	Local Name
Assam, Meghalaya, Mizoram, Nagaland	Jhumming
Manipur	Pamlou
Bastar (Chhattisgarh) & A&N Islands	Dipa
Madhya Pradesh	Bewar / Dahiya
Andhra Pradesh	Podu / Penda
Odisha	Pama Dabi / Koman / Bringa
Western Ghats	Kumari
South-eastern Rajasthan	Valre / Waltre
Himalayan belt	Khil
Jharkhand	Kuruwa

3. PRIMITIVE SUBSISTENCE FARMING

- This type of farming is still practised in a **few pockets of India**.
- Practised on small patches of land with the help of:
 - primitive tools like **hoe, dao and digging sticks**
 - **family/community labour**
- Depends upon: **monsoon, natural fertility** of soil, and suitability of environmental conditions.

International Names for Slash & Burn

Country / Region	Name
Mexico and Central America	Milpa
Venezuela	Conuco
Brazil	Roca
Central Africa	Masole
Indonesia	Ladang
Vietnam	Ray

4. RINJHA — EXAMPLE OF PRIMITIVE SUBSISTENCE

- Rinjha lived with her family in a small village in the outskirts of **Dhipu in Assam**.
- Watched family members clearing, slashing, and burning land for cultivation.
- Irrigated fields with water running through a **bamboo canal** from a nearby spring.
- **Takeaway:** Rinjha's family is engaged in primitive subsistence / shifting cultivation.

5. INTENSIVE SUBSISTENCE FARMING

- Practised in areas of **high population pressure** on land.
- It is **labour-intensive farming**.
- High doses of biochemical inputs and irrigation are used for higher production.

6. COMMERCIAL FARMING

- Main characteristic: Use of **higher doses of modern inputs** (HYV seeds, chemical fertilisers, insecticides, pesticides) to obtain higher productivity.
- Degree of commercialisation varies by region:
 - **Rice** is a commercial crop in Haryana & Punjab, but a subsistence crop in Odisha.

7. PLANTATION FARMING

- A single crop is grown on a large area.
- Interface of **agriculture and industry**.
- Uses capital intensive inputs with migrant labourers.
- **Important Crops:** Tea, Coffee, Rubber, Sugarcane, Banana.
- **Key Areas:** Tea in Assam & North Bengal; Coffee in Karnataka.
- Requires a well-developed network of transport and communication to connect plantations, processing units, and markets.

Why enormous pressure on land?

The '**right of inheritance**' leads to division of land among successive generations, rendering land-holding size uneconomical. Absence of alternative livelihoods forces farmers to seek maximum output.

8. CROPPING PATTERN

India has three main cropping seasons: **Rabi**, **Kharif**, and **Zaid**.

Season	Sowing Time	Harvesting Time	Important Crops	Major Regions / Key Factors
RABI	Winter (Oct–Dec)	Summer (Apr–Jun)	Wheat, barley, peas, gram, mustard	Punjab, Haryana, HP, J&K, Uttarakhand, UP. Benefits from winter western temperate cyclones & Green Revolution.
KHARIF	Onset of monsoon	Sep–Oct	Paddy, maize, jowar, bajra, tur, moong, urad, cotton, jute, groundnut, soyabean	WB, Assam, Odisha, AP, Telangana, TN, Kerala, Konkan coast, UP, Bihar, Punjab, Haryana.
ZAID	Summer (between Rabi & Kharif)	Summer months	Watermelon, muskmelon, cucumber, vegetables, fodder crops	Short season. Sugarcane takes almost a year to grow.

Paddy in East India: In Assam, West Bengal, and Odisha, three crops of paddy are grown in a year: **Aus**, **Aman**, and **Boro**.

9. MAJOR CROPS — FOOD GRAINS

RICE

- Staple food crop; India is **2nd largest producer** after China.
- **Kharif crop**. Requires temp **above 25°C**, high humidity, annual rainfall **above 100 cm**.
- Grown in plains of north/north-eastern India, coastal areas, deltaic regions.

WHEAT

- 2nd most important cereal crop; main food crop in North & North-West.
- **Rabi crop**. Requires cool growing season, bright sunshine at ripening, **50–75 cm** rainfall.

MILLETS (Coarse Grains)

- **Jowar**: 3rd most important food crop. Rain-fed; grown in moist areas (MH, Karnataka, AP, MP).
- **Bajra**: Grows on sandy and shallow black soil (Rajasthan, UP, MH, Gujarat, Haryana).
- **Ragi**: Rich in **iron**, **calcium**, **roughage**. Dry region crop (Karnataka, TN, HP, Uttarakhand, Sikkim, Jharkhand, AP).

MAIZE & PULSES

- **Maize**: Food & fodder crop; Kharif (Rabi in Bihar); temp 21°C–27°C; old alluvial soil.

- **Zones:** Ganga-Satluj plains, Deccan black soil region. (Punjab, Haryana, UP, MP, Bihar, Rajasthan).

- **Pulses:** India is largest producer & consumer; major protein source. Leguminous (fix nitrogen). Needs less moisture. (MP, Rajasthan, MH, UP, Karnataka).

10. FOOD CROPS OTHER THAN GRAINS

SUGARCANE

- Tropical/subtropical; 2nd largest producer after Brazil.
- Temp: 21°C–27°C, Rainfall: 75–100 cm. Needs manual labour.
- Source of sugar, gur, khandsari, molasses. (UP, MH, Karnataka, TN).

OIL SEEDS (12% of cropped area)

- **Groundnut:** Kharif crop; half of oilseeds produced. Gujarat is top producer.
- **Linseed & Mustard:** Rabi crops.
- **Sesamum:** Kharif in North; Rabi in South.
- **Castor:** Grown both as Rabi and Kharif.

TEA & COFFEE

- **Tea:** Tropical/subtropical, deep fertile well-drained soil rich in humus. Labour-intensive. (Assam, WB, TN, Kerala). 2nd largest producer after China.
- **Coffee:** Arabica variety (from Yemen). Grown in Baba Budan Hills, Nilgiris (Karnataka, Kerala, TN).

11. NON-FOOD & FIBRE CROPS

RUBBER

- Equatorial crop; moist/humid climate, rainfall > 200 cm, temp > 25°C.
- Kerala, TN, Karnataka, A&N Islands, Garo Hills (Meghalaya).

FIBRE CROPS & SERICULTURE

- Four major fibres: Cotton, Jute, Hemp, Natural Silk.
- **Sericulture:** Rearing silkworms on mulberry leaves for silk.
- **Cotton:** Black soil, high temp, light rainfall, **210 frost-free days**, bright sunshine. 6-8 months to mature. (MH, Gujarat, MP).
- **Jute ("Golden Fibre"):** Fertile flood plain soils, high temp. Used for gunny bags, mats, ropes. (West Bengal, Bihar, Assam, Odisha, Meghalaya).

12. TECHNOLOGICAL & INSTITUTIONAL REFORMS

Institutional Reforms (Post-Independence)

- Collectivisation, consolidation of holdings, cooperation, abolition of zamindari.
- Land reform was main focus of the First Five Year Plan.

1960s & 1970s Strategies

- **Green Revolution:** Based on package technology (HYV seeds, fertilisers).
- **White Revolution:** (Operation Flood) to increase milk production.

1980s & 1990s Comprehensive Programme

- **Crop Insurance** against drought, flood, cyclone, fire, disease.
- Establishment of Grameen banks & cooperative societies for low-interest loans.

Government Schemes & Measures

- **Kisan Credit Card (KCC)** and **Personal Accident Insurance Scheme (PAIS)**.
- Special weather bulletins & agricultural broadcasts on radio/TV.
- **Minimum Support Price (MSP)**, remunerative, and procurement prices to stop exploitation by middlemen/speculators.

13. BHOODAN – GRAMDAN

- Initiated by **Vinoba Bhave** (spiritual heir of Mahatma Gandhi).
- **Bhoodan:** At Pochampally (Telangana), **Shri Ram Chandra Reddy** offered 80 acres of land to 80 landless villagers.
- **Gramdan:** Zamindars offered villages to landless.
- Movement is also known as the **Blood-less Revolution**.